

Scout Report — Phase 21 Canonical PostgreSQL Data Model Completion & Verification

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1. Phase objective

Convert the Phase 20 approved 93-field canonical data dictionary into a backward-compatible PostgreSQL model using the required collect → inventory → map → approve → code sequence.

2. Collect

Phase 20 source inventory and canonical dictionary were used as the authoritative input. Existing Scout Report schema and the four-stage verification workflow were preserved.

3. Inventory

The model covers identity, location, crop, visit, weather, soil, water, weed, pest, disease, nutrient, stress, management, recommendations, evidence, diagnostics and import provenance.

4. Map

Report-level fields were added to scout_reports. Repeatable field observations were normalized into dedicated child tables and linked to report_survey_stops where sampling location matters.

5. Approve

The Phase 20 dictionary status is phase20-approved-for-phase21. No canonical field was silently removed. Import validation remains separate from workflow approval.

6. Code

server/migrations/004_expanded_scouting_model.sql is transactional and idempotent for the implemented DDL. It adds 13 report-header columns, 14 child/provenance tables, constraints and indexes, and extends existing pest/disease tables.

7. Verification

Static Phase 21 verifier: PASS. Canonical model unit tests: 2/2 PASS. Full npm test attempt was blocked by missing dependencies in the execution environment. Live PostgreSQL migration was not claimed as passed because a configured database runtime was unavailable.

8. Exact next environment gate

Run npm ci, npm run verify:phase21, npm run migrate, then npm test against the configured PostgreSQL environment. Only after that should Phase 22 API/services and dynamic form work consume the new schema.

9. Next measure

Phase 22 — API/services and dynamic comprehensive Scout Report form backed by the canonical model, followed by expanded import/export and analytics integration.

Verification metrics

Metric	Result
Canonical fields	93
New child/provenance tables	14
Report header extensions	13
Static verifier	PASS
Canonical model tests	2/2 PASS
Full application suite	BLOCKED: dependencies
Live PostgreSQL migration	PENDING configured DB